Extended Abstract

Title: Explainable Deep Learning Approach for Sleep Disorder Detection and Personalized Lifestyle Recommendations

Research Summary: This work presents an AI-driven Sleep Disorder Management System that performs automated sleep staging, multi-disorder detection, interpretable reporting, and personalized recommendation. Building upon SOTA sleep staging architectures, the study systematically experiments with advanced modeling techniques and training strategies to achieve competitive performance and deployability. Since accurate staging forms the foundation for downstream clinical variable extraction, it enables medically meaningful disorder detection. To reduce black-box complexity, multiple machine learning models are evaluated for performance and inference efficiency, identifying suitable architectures for each disorder.

Objective of the study: Despite advances in automated sleep staging and disorder detection, existing approaches rely heavily on using deep neural networks with inherent black-box nature, lacking interpretability and clinical transparency[1]. Most systems focus solely on classification accuracy and do not integrate disorder management or patient interaction[2]. Furthermore, multimodal PSG-based models increase hardware complexity and inference cost[3]. This study addresses these gaps by proposing a unified end-to-end framework that integrates automated staging, multi-disorder risk prediction, explainable AI, and personalized management. The key contributions include a lightweight focused architecture, integration of clinically validated features for disorder detection, expansion toward underexplored disorders, and a clinically interpretable decision-support ecosystem.

Methodology: The proposed framework consists of automated sleep staging followed by disorder detection and management modules, as illustrated in Figure 1. The staging architecture follows a hierarchical temporal-spectral fusion design. At the epoch level, raw EEG signals are processed using adaptive atrous convolutional pyramids to capture multi-scale temporal and frequency-sensitive patterns. Squeeze-and-excitation blocks recalibrate channel responses to emphasize informative EEG components. In parallel, deterministic spectral features derived from wavelet and STFT-based transformations encode complementary frequency-domain characteristics. An attention-based fusion mechanism integrates temporal and spectral representations, generating compact embeddings for each epoch. These embeddings are then passed to a transformer encoder with positional encoding to model long-range inter-epoch dependencies and structured sleep transitions through self-attention.

For disorder detection, automated staging outputs are converted into clinically validated sleep architecture features and used to classify insomnia, sleep apnea, restless leg syndrome, and related disorders using multiple machine learning models. An explainability layer and clinical reporting module complete the end-to-end system.

Results: The proposed system was evaluated on Sleep-EDF (193 subjects) and validated using the Multi-Ethnic Study of Atherosclerosis dataset. The model achieved 79.52% accuracy for five-class sleep staging with a macro-average AUROC of 0.978 showing state-of-the-art performance in detecting N1 and REM sleep, this proves that our model is highly competitive with the SOTA models especially with the N1 stage performance which is always lowest scored due to the lack of distinctive characteristics. Apnea detection achieved an AUROC of 75.7%, insomnia 62.9%, and restless leg syndrome 62.2%.

Figure 2 presents the overall performance comparisons across staging.

Table 1 provides a comparative analysis against recent state-of-the-art models, including transformer-based sleep staging frameworks and temporal spectral fusion models. While some existing works report higher staging accuracy, the proposed system uniquely integrates lightweight design, interpretability and multi-disorder detection, and patient-facing management within a single scalable framework. This holistic integration distinguishes it from prior isolated staging or classification approaches.

Method	**Per class F1 Score**					**Weighted Average Accuracy%**
	N1 %	**N2 %**	**N3 %**	**REM %**	**Wake %**	
MultiSEss [4]	47.62*	83.51*	75.36*	75.11*	91.58*	82.30
ADG SleepNet [5]	53.7	**90.0**	**85.5**	84.8	91.3	**85.1**
BiTS SleepNet [6]	50.23	87.05	82.50	85.33	93.33	85.09
TSA-Net [7]	30.11	84.94	80.02	81.03	91.71	82.21
Sleep Transformer [8]	48.5	86.5	80.9	84.6	**93.5**	84.9
Temporal Spectral Cross Attention Fusion Transformer (ours)	**59.98**	76.27	74.79	**89.25**	91.13	80.67

Table I: Comparison among proposed and SOTA models. The numbers in bold indicate the best performance, and the numbers underlined are the second best. * - The paper didn't record the per class f1 scores instead per class precision is provided thus it is incomparable to the others.

Method	**Apnea Detection Performance**	**Insomnia Detection Performance**	**PLMB Detection Performance**
Faust et al., 2021	99.80%	-	-
Sharma et al., 2022	-	99.70%	97.50%
Almuhammadi et al., 2015	97.14%	-	-
Atianashie et al., 2021	-	89%	-
Ours	75.7 (Random Forest)	62.9 (Linear Regression)	62.2 (SVM with RBF kernel)

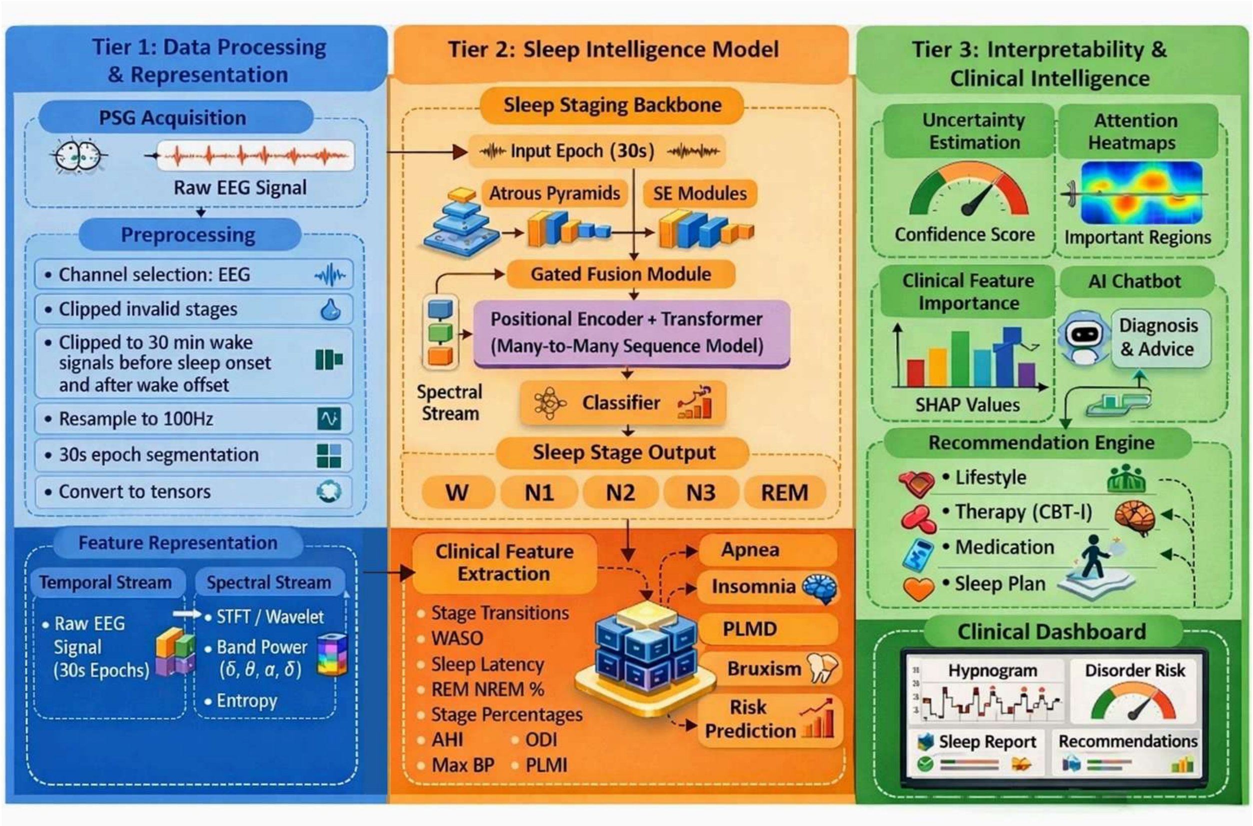

Figure 1: Data flow of our proposed system, modularized into three tiers. The data uploaded by the patient will undergo processing for noise removal and better representation(on the left), further the transformed data is fed to the sleep staging backbone to sequence the sleep stages (top middle) and clinically used features are extracted from the sequence and fed to various ML classifier (bottom middle), Finally the predictions from the models are collected and various interpretable results will be represented as clinically meaningful results and provides assistive features to layman patients (on the right).

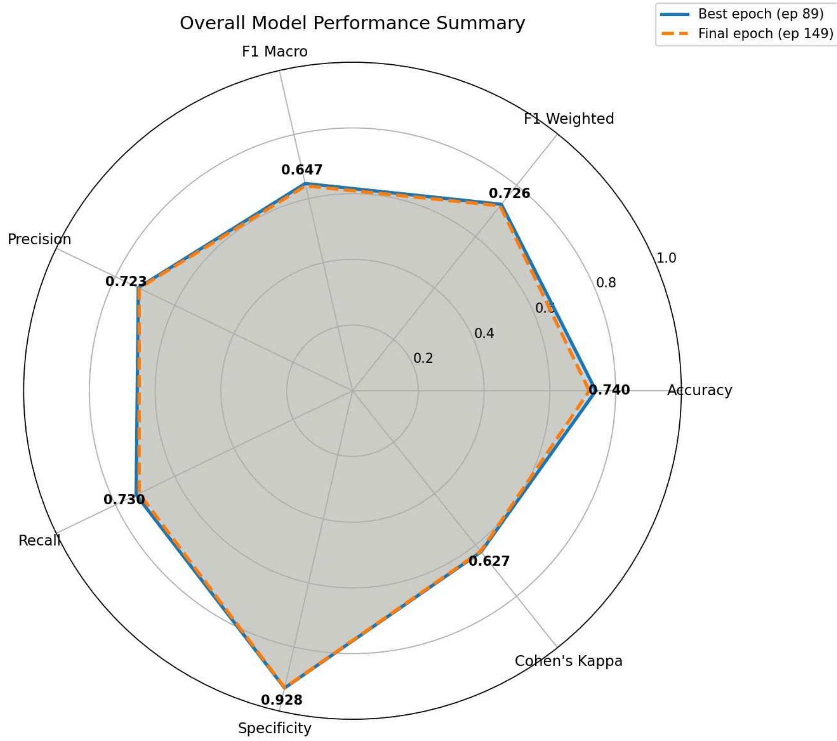

Figure 2: This figure presents the overall performance of the sleep staging backbone in various evaluation metrics which can be directly compared and benchmarked with existing SOTA models.

References

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